

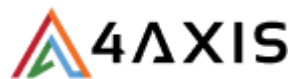
4Axis: Inspiring creativity with a seamless, AI- enhanced drawing app on Firebase

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ABOUT 4AXIS

4Axis scales its [Drawing] Desk app using Firebase without managing infrastructure, and user data using Elasticsearch for actionable insights. It accelerates AI development with Vertex AI.



About 4Axis

4Axis Solutions, founded in 2012, creates drawing apps for professionals and general users. With over 50 million downloads, Drawing Desk is one of the most popular drawing apps in the App Store and Google Play. 4Axis continuously innovates to deliver new features, such as AI colorization and interactive tutorials that enhance the user's drawing experience. Over the years, 4Axis has helped families in need, hospitals, and schools through its corporate social responsibility initiatives.

Google Cloud results

- Stores 3TB of user-created content in Cloud Storage for secure and fast retrieval
- Scales to 200,000 concurrent users on Firebase for a seamless drawing experience
- Analyzes 1.75TB of Google Analytics data with BigQuery to map user journeys

Connected million zero d

Industries: Technology

Location: Sri Lanka

(<https://firebase.google.com/docs/crashlytics/>),
Google Cloud (<https://cloud.google.com/>) ex-ai/)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Storage (<https://cloud.google.com/storage/>)

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Firebase (<https://firebase.google.com/>)

Firebase Crashlytics
(<https://firebase.google.com/docs/crashlytics/>)

Vertex AI (<https://cloud.google.com/vertex-ai/>)

People create art for a variety of reasons. While some people sketch for self-expression, others color for relaxation. 4Axis (<https://4axis.com/>), an app developer, transforms the creative journey with its Drawing Desk app, so anyone can use their mobile device as a canvas when inspiration strikes. With a wide selection of digital art tools, including color editing and unwanted object removal, Drawing Desk is popular with both adults and children.

In 2016, 4Axis' legacy architecture was strained under the pressure of heavy traffic, with daily downloads increasing five-fold from 3,000 to 15,000 year-on-year. Users experienced frequent downtime due to server crashes. When such problems occurred, the lean team of two developers had to stop coding and scramble to increase RAM.

"We're focused on the user. Firebase provides a scalable and stable architecture that delivers zero downtime, without us having to make server changes repeatedly," says Dumindu Kanankage, Co-Founder and Chief Executive Officer at 4Axis. "For a smooth user experience, our managed backend automatically scales with traffic and expands to 200,000 concurrent users."



Optimizing the user experience by syncing data in real-time between apps and devices

4Axis migrated to Google Cloud in 2017 because

Firestore Realtime Database

(<https://firebase.google.com/docs/database>) was the only option for storing and syncing real-time data from users. Firestore managed services allow developers to focus on app development, such as AI features in Drawing Desk. Apart from the Realtime Database, 4Axis also uses Cloud Storage

(<https://cloud.google.com/storage>) to store user content, Remote Config

(<https://firebase.google.com/docs/remote-config>) to customize the app, and Crashlytics

(<https://firebase.google.com/products/crashlytics>) to monitor app stability.

Firestore Realtime Database is the main component of the Drawing Desk application, which connects more than 50 million users. 4Axis hosts 3TB of user-generated content in Cloud Storage and saves file paths unique to each user in the Realtime Database. When a user shares a project with the community, the app retrieves the file path so everyone can view

the artwork immediately. If the user goes offline, the app automatically syncs changes with Firebase when they return online.

In order to build customer loyalty and trust, it is crucial to secure user data. Using Firebase Security Rules, 4Axis protects data in Realtime Database and Cloud Storage by defining who can access what data. As a result, users can create and share artwork with confidence.

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Co-Founder and Chief
Executive Officer, 4Axis*

Enhancing app engagement with actionable insights from BigQuery

Drawing Desk has a loyal following and continues to attract new customers after a decade. Adapting to changing customer preferences ensures the app's longevity. Therefore, developers must know how users interact with the app and what features they prefer.

"Using Google Analytics, the app gathers data from user events such as app launches and content viewings. With BigQuery, we analyze 1.75TB of app data across different devices to gain a deeper understanding of user journeys," says Kanankage. "Based on historical data, we can determine the most effective way to convert trial users into paid

subscribers, such as premium features and discounts."

Today, 4Axis continues to optimize its use of Google Cloud by integrating Firebase with [BigQuery](https://cloud.google.com/bigquery) (<https://cloud.google.com/bigquery>) for data analytics and [Vertex AI](https://cloud.google.com/vertex-ai) (<https://cloud.google.com/vertex-ai>) for machine learning.

Attracting a global audience with Remote Config for localized features

AI colorization is one of the premium features driving app engagement and conversion. Based on user feedback, 4Axis developed a machine learning algorithm to help users complete their sketches and colorize their artwork in seconds. With an open-source dataset of 300,000 Anime sketches, 4Axis trained the AI feature to predict what users are drawing and help them complete it. With Google Cloud, 4Axis can run compute-intensive data pipelines, such as machine learning model training and prediction, without impacting app performance.

The team can also measure user engagement before launching new features, by using Remote Config for A/B testing to release new changes to different user groups. Similarly, 4Axis uses Remote Config to localize app content for its global audience.

The localized content, such as language and user interface, is available to existing users without installing a new version from the App Store or

Google Play. Users in the U.S. can, for instance, see a sticker pack for Thanksgiving, while users in India can find one for Diwali.

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Achieving app stability with Crashlytics realtime crash reports

It is common for users to become frustrated and uninstall an app or leave poor reviews if it crashes. Crashlytics' real-time crash reports help 4Axis monitor and debug stability issues. Based on severity, Crashlytics sends alerts to developers and the executive team via a Jira IT ticketing system.

"Our team uses Crashlytics to identify when and why crashes occur, so we can spend less time troubleshooting and focus on fixing the problem faster," says Kanankage. "With Crashlytics, we can track, prioritize, and resolve issues to improve app performance."

In another premium feature, 4Axis is exploring AI to improve users' drawing skills with step-by-step tutorials. Instead of watching a training video, users can learn and practice digital art techniques in the app, such as drawing line art and using opacity masks. By recommending lessons that are appropriate for the user's level, the AI feature engages the user.

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Building a cloud talent pool for future growth

Following the success of Drawing Desk, one of the highest-ranked digital art apps in the App Store and Google Play Store, 4Axis plans to launch new apps and expand its global footprint in the next five to ten years. But for the company to continue growing, it also needs to hire more talent.

"Having been a Google Strategic Partner with AdMob since our inception, we have had a close working relationship with Google. Recruits and graduates can get hands-on experience using the

latest Google Cloud products and learn best practices from the Google team," says Kanankage.

"Thanks to Google Cloud, we offer a positive environment for newcomers to boost their cloud skills and grow with the company."